

12M3. As with the zero-inflated Poisson, the zero-inflated binomial mixes some extra zeros into another distribution. The structure is very much the same as the ZI-Poisson. First, a single probability determines whether or not a zero is observed. Call this probability p_0 . When a zero is not observed from this first process, the binomial distribution takes over. It may also generate a zero. Call the probability of a success from the binomial process q , and let it have n trials. Then the probability of a zero, mixing together both processes, is:

$$\Pr(0|p_0, q, n) = p_0 + (1 - p_0)(1 - q)^n$$

The logic is that either we get a zero from the first process, p_0 of the time, or we got a zero from the binomial, which happens only when all trials fail, $(1 - q)^n$ of the time. The probability of any particular non-zero observation y is similarly:

$$\Pr(y|p_0, q, n) = (1 - p_0) \frac{n!}{y!(n - y)!} q^y (1 - q)^{n - y}$$

Compare this expression to the ZI-Poisson expression in the chapter.

12H1. The problem left you freedom to specify your own priors. We'll need to do some prior simulations again, of course. This means thinking about typical ranges of hurricane deaths (the intercept) and how much of an effect the name could possibly have (the slope).

Let's start with a very simple Poisson model predicting deaths using only femininity as a predictor. I'm going to standardize femininity, for the usual reasons. Then we'll simulate from the priors and see how not to make the prior predictions silly.

R code
12.6

```
library(rethinking)
data(Hurricanes)
d <- Hurricanes
d$fmnty_std <- ( d$femininity - mean(d$femininity) )/sd(d$femininity)
dat <- list( D=d$deaths , F=d$fmnty_std )

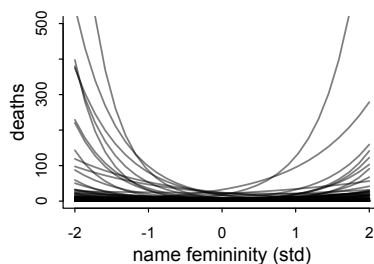
# model formula - no fitting yet
f <- alist(
  D ~ dpois(lambda),
  log(lambda) <- a + bF*F,
  a ~ dnorm(1,1),
  bF ~ dnorm(0,1) )
```

Now if you aren't an expert on hurricanes, you won't have a sense of what the plausible range of D should be. The most deadly hurricane in recorded history is the 1900 "Great Galveston hurricane," which killed somewhere between 6000 and 10000 people. Note that this storm doesn't have a personal name, and it doesn't appear in the sample, because storms didn't get names until after 1950. Most storms since 1950 are also much less deadly than this, killing fewer than a dozen people. But occasionally there are storms that kill a hundred or more. Hurricane mortality is a thick-tailed phenomenon. So you might have a hunch already that a simple Poisson model is not going to do a good job. But let's do our best. What do the priors above produce?

R code
12.7

```
N <- 100
a <- rnorm(N,1,1)
bF <- rnorm(N,0,1)
F_seq <- seq( from=-2 , to=2 , length.out=30 )
plot( NULL , xlim=c(-2,2) , ylim=c(0,500) ,
      xlab="name femininity (std)" , ylab="deaths" )
```

```
for ( i in 1:N ) lines( F_seq , exp( a[i] + bF[i]*F_seq ) , col=grau() , lwd=1.5 )
```



This allows for some rather implausible extreme trends. But the typical trend is very modest, only a very small increase (or decrease) in deaths as femininity of the name changes. I'd rather use a tighter prior on the slope, because it makes no sense to think that the femininity of a storm's name could increase the deaths by an order of magnitude. But let's proceed with these priors and see what happens.

To fit the model:

```
m1 <- ulam( f , data=dat , chains=4 , log_lik=TRUE )
precis( m1 )
```

R code
12.8

```
      mean    sd 5.5% 94.5% n_eff Rhat4
a   3.00  0.02  2.96  3.04  1219    1
bF  0.24  0.03  0.20  0.28  1069    1
```

The model seems to think there is a reliable association between femininity of name and deaths.

Now in order to see which hurricanes the model retrodicts well, we can plot the implied trend over the raw data points. I'll compute and plot the expected death count, 89% interval of the expectation, and 89% interval of the expected distribution of deaths (using Poisson sampling).

```
# plot raw data
plot( dat$F , dat$D , pch=16 , lwd=2 ,
      col=rangi2 , xlab="femininity (std)" , ylab="deaths" )

# compute model-based trend
pred_dat <- list( F=seq(from=-2,to=1.5,length.out=30) )
lambda <- link( m1 , data=pred_dat )
lambda.mu <- apply(lambda,2,mean)
lambda.PI <- apply(lambda,2,PI)

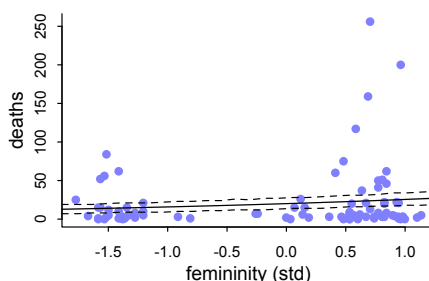
# superimpose trend
lines( pred_dat$F , lambda.mu )
shade( lambda.PI , pred_dat$F )

# compute sampling distribution
deaths_sim <- sim(m1,data=pred_dat)
deaths_sim.PI <- apply(deaths_sim,2,PI)

# superimpose sampling interval as dashed lines
lines( pred_dat$F , deaths_sim.PI[1,] , lty=2 )
lines( pred_dat$F , deaths_sim.PI[2,] , lty=2 )
```

R code
12.9

This is the result:



We can't even see the 89% interval of the expected value, because it is so narrow. The sampling distribution isn't much wider itself. What you can see here is that femininity accounts for very little of the variation in deaths, especially at the high end. There's a lot of over-dispersion, which is very common in Poisson models. As a consequence, this homogenous Poisson model does a poor job for most of the hurricanes in the sample, as most of them lie outside the prediction envelop (the dashed boundaries).

Any trend also seems to be driven by a small number of extreme storms with feminine names. As you might expect, PSIS does not like this:

```
R code
12.10 stem( PSISk( m1 ) )
```

Some Pareto k values are very high (>1)

The decimal point is 1 digit(s) to the left of the |

```
-0 | 8521
0 | 00011223333455566789999000001111111122233444455556666666777778899
2 | 000011122289
4 | 1
6 | 87
8 | 59
10 |
12 | 9
14 | 9
16 | 8
18 |
20 |
22 | 3
```

There are four storms with k values above 1, and one above 2!

12H2. To deal with the over-dispersion seen in the previous problem, now we'll fit a Poisson model with varying rates, a gamma-Poisson model. This code will set up the data (just as in the previous problem) and fit the gamma-Poisson model:

```
R code
12.11 library(rethinking)
data(Hurricanes)
d <- Hurricanes
d$fmnty_std <- ( d$femininity - mean(d$femininity) )/sd(d$femininity)
dat <- list( D=d$deaths , F=d$fmnty_std )
```

```
m2 <- ulam(
  alist(
    D ~ dgamma( lambda , scale ),
    log(lambda) <- a + bF*F,
    a ~ dnorm(1,1),
    bF ~ dnorm(0,1),
    scale ~ dexp(1)
  ), data=dat , chains=4 , log_lik=TRUE )
```

Inspect the marginal posterior distributions of the parameters:

```
precis(m2)
```

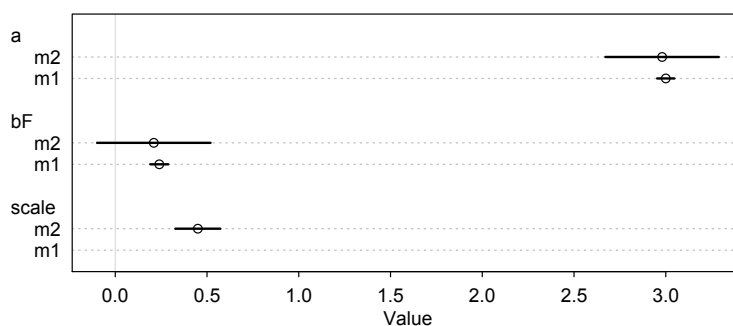
R code
12.12

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a	2.98	0.16	2.73	3.23	1597	1
bF	0.21	0.16	-0.05	0.46	1435	1
scale	0.45	0.06	0.35	0.56	1652	1

Note that the 89% interval for bF now overlaps zero, because it has more than 5 times the standard deviation as the analogous parameter in model m1. For a head-to-head comparison, let's do a graphical `coefstab`:

```
plot(coefstab(m1,m2))
```

R code
12.13



Model m2 has nearly the same posterior means for the intercept and slope bF, but much more uncertainty.

How does this translate into predictions? To find out, let's do the plotting from the previous problem over again, using model m3 now:

```
# plot raw data
plot( dat$F , dat$D , pch=16 , lwd=2 ,
      col=range(2 , xlab="femininity (std)" , ylab="deaths" ) )

# compute model-based trend
pred_dat <- list( F=seq(from=-2,to=1.5,length.out=30) )
lambda <- link(m2,data=pred_dat)
lambda.mu <- apply(lambda,2,mean)
lambda.PI <- apply(lambda,2,PI)

# superimpose trend
```

R code
12.14

```

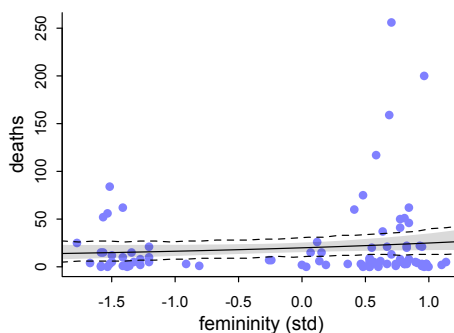
lines( pred_dat$F , lambda.mu )
shade( lambda.PI , pred_dat$F )

# compute sampling distribution
deaths_sim <- sim(m2,data=pred_dat)
deaths_sim.PI <- apply(deaths_sim,2,PI)

# superimpose sampling interval as dashed lines
lines( pred_dat$F , deaths_sim.PI[1,] , lty=2 )
lines( pred_dat$F , deaths_sim.PI[2,] , lty=2 )

```

This is the new result:



There is more uncertainty now about the relationship, and the prediction interval is wider. But the predictions are still terrible.

Now for the conceptual part of this problem: Why does including varying rates, via the gamma distribution, result in greater uncertainty in the relationship? The gamma-Poisson model allows each hurricane to have its own unique expected death rate, sampled from a common distribution that is a function of the femininity of hurricane names. We can actually plot this distribution from the posterior distribution, for any given femininity value. I'll produce three examples, plotting 100 randomly sampled gamma distributions of the rate of deaths for three different femininity values:

R code
12.15

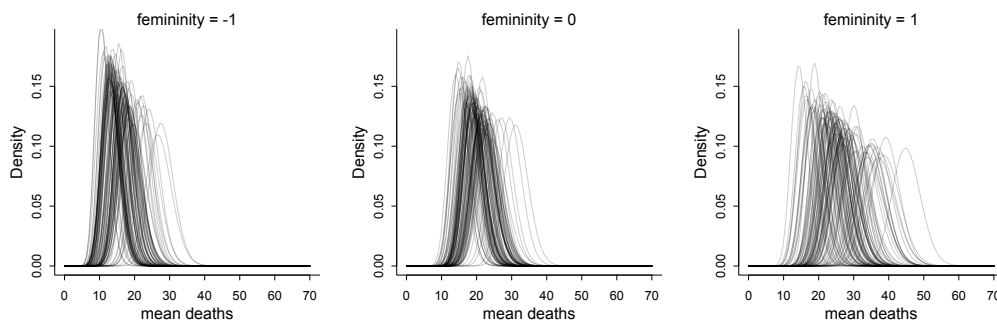
```

post <- extract.samples(m2)

fem <- (-1) # 1 stddev below mean
for ( i in 1:100 )
  curve( dgamma2(x,exp(post$a[i]+post$bF[i]*fem),post$scale[i]) ,
         from=0 , to=70 , xlab="mean deaths" , ylab="Density" ,
         ylim=c(0,0.19) , col=col.alpha("black",0.2) ,
         add=ifelse(i==1,FALSE,TRUE) )
mtext( concat("femininity = ",fem) )

```

And just change the value assigned to fem above to make the other plots. Here are the plots:



Each gray curve above is a gamma distribution of mean death rates, sampled from the posterior distribution of the model. This is an inherently confusing thing: a distribution sampled from a distribution. So let's take it again, slowly. A gamma distribution is defined by two parameters: a mean and a scale. The mean in this model is controlled by the linear model and its two parameters, a and b_f . The code above takes single values of a and b_f from the posterior distribution and builds a single linear model. It's exponentiated in the code, because this model uses a log link. And the parameter `scale` is the scale, so you can see that in the code as well. 100 gamma distributions are drawn for each given value of femininity. This visualizes the uncertainty in the posterior about the variation in death rates. Read that again, slowly. It is a bit weird, but with a little time, it makes plenty of sense. Just like a simple parameter like an intercept has uncertainty, and the posterior distribution measures it (given a model and data), a function of parameters like a gamma distribution will also have uncertainty. Essentially there are an infinite number of gamma distributions that are possible, the the Bayesian model has considered all of them and ranked them by their plausibility. Each plot above shows 100 such gamma distributions, sampled from the posterior distribution in proportion to their plausibilities.

So now back to the explanation of why the parameters a and b_f have wider posterior distributions. Once we allow any given values of a and b_f to produce many different death rates, because they feed into a gamma distribution that produces variation, then many more distinct values of a and b_f can be consistent with the data. This results in wider posterior distributions. The same phenomenon will reappear when we arrive at multilevel models in Chapter 13.

You might be curious how `m2` compares to `m1`, in terms of PSIS/WAIC. If so, take a look. You'll find that the effective number of parameters for `m1` is very very large. Adding a parameter to `m2` actually makes the model less prone to overfitting. Can you explain why?

A later problem gets at that.

12H6. Read in the data:

```
library(rethinking)
data(Fish)
d <- Fish
str(d)
```

R code
12.26

```
'data.frame': 250 obs. of 6 variables:
 $ fish_caught: int  0 0 0 0 1 0 0 0 0 1 ...
 $ livebait   : int  0 1 1 1 1 1 1 1 0 1 ...
 $ camper     : int  0 1 0 1 0 1 0 0 1 1 ...
 $ persons    : int  1 1 1 2 1 4 3 4 3 1 ...
 $ child      : int  0 0 0 1 0 2 1 3 2 0 ...
 $ hours      : num  21.124 5.732 1.323 0.548 1.695 ...
```

I gave you to have a lot of freedom here. So I'm not expecting any particular model. But I do expect you to have constructed a correct zero-inflated Poisson (ZIPoisson) model and use the exposure/offset correctly. So here's an example, using a subset of the predictors:

```
d$loghours <- log(d$hours)
m12H6a <- ulam(
  alist(
    fish_caught ~ dzipois( p , mu ),
    logit(p) <- a0 + bp0*persons + bc0*child,
```

R code
12.27

```
log(mu) <- a + bp*persons + bc*child + loghours,
c(a0,a) ~ dnorm(0,1),
c(bp0,bc0,bp,bc) ~ dnorm(0,0.5)
) , data=d , chains=4 )
```

There are two key things to note. First, the parameters are different in the two linear models. No parameters are shared among them. This allows the same predictors to potentially influence each part of the process in different ways. Second, the offset `loghours` is added onto the end of the linear model for `log(mu)`, the mean of the Poisson process producing fish. This corrects for the fact that some visitors stayed longer, so had more opportunity to catch fish.

Here are the marginal posterior distributions from the model above:

R code
12.28

```
precis(m12H6a)
```

	mean	sd	5.5%	94.5%	n_eff	Rhat4
a	-2.25	0.14	-2.49	-2.04	1159	1
a0	0.41	0.46	-0.34	1.12	1455	1
bc	0.53	0.09	0.38	0.67	1327	1
bp	0.66	0.04	0.60	0.73	1152	1
bc0	0.50	0.46	-0.28	1.20	1472	1
bp0	-0.76	0.21	-1.10	-0.43	1460	1

Note that the parameters `a0`, `bp0`, and `bc0` are on the logit (log-odds) scale, while the parameters `a`, `bp`, and `bc` are on the log scale.

Here's an example of generating predictions. We need to consider each component of the process separately, and actually `link` will respect this, because it computes the inverse-link values for all linear models in the formula list you provided:

R code
12.29

```
zip_link <- link(m12H6a)
str(zip_link)
```

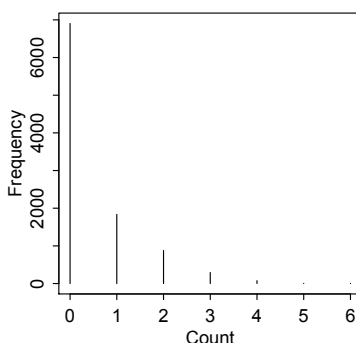
```
List of 2
 $ p : num [1:2000, 1:250] 0.441 0.345 0.459 0.503 0.351 ...
 $ mu: num [1:2000, 1:250] 4.38 4.46 4.01 4.37 4.34 ...
```

The `p` matrix is probabilities of not fishing for each sample/case, and the `mu` matrix is the expected number of fish caught (conditional on fishing) for each sample/case.

So what do these matrices imply about posterior predictions? You have to use the model, as always. The `p` values provide predictions for excess zeros, and the `mu` values for the Poisson process (which can also produce zeros). The an observation requires using both. For example, if $p = 0.5$ and $\mu = 1$ (fixed right now for ease of understand), then the implied predictive distribution is:

R code
12.30

```
zeros <- rbinom(1e4,1,0.5)
obs_fish <- (1-zeros)*rpois(1e4,1)
simplehist(obs_fish)
```

Now luckily `sim` understands all of this, because it knows the model already. So you can just:

```
fish_sim <- sim(m12H6a)
str(fish_sim)
```

R code
12.31

```
num [1:1000, 1:250] 0 0 0 0 0 0 4 1 1 0 ...
```

And these simulations integrate over the posterior uncertainty, so there is one simulation for each sample.

You could go on to summarize and plot these simulations. But what we actually want is counterfactual posterior predictions. So let's assume for example a party of 1 person spending 1 hour in the park. Let's do that now:

```
# new data
pred_dat <- list(
  loghours=log(1), # note that this is zero, the baseline rate
  persons=1,
  child=0 )

# sim predictions - want expected number of fish, but must use both processes
fish_link <- link( m12H6a , data=pred_dat )

# summarize
p <- fish_link$p
mu <- fish_link$mu
( expected_fish_mean <- mean( (1-p)*mu ) )
( expected_fish_PI <- PI( (1-p)*mu ) )
```

R code
12.32

```
[1] 0.1191649
```

```
      5%      94%
0.0925886 0.1471911
```

What this means is that one person is expected to extract on average 0.12 fish per hour, with the interval shown. This estimate allows for the zero-inflation, so it accounts for the probability that the person is not fishing at all, depressing the expected value.

12H7. This is my DAG: